

## electrostatics - practice problems

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1. A  $+3.0 \mu\text{C}$  charge is  $0.40 \text{ m}$  to the left of a  $+2.0 \mu\text{C}$  charge. What are the magnitude and direction of the force on the left charge?
2. Two charged objects exert a  $0.80 \text{ N}$  force on each other when they are  $0.20 \text{ m}$  apart. If the same charges are moved to  $0.40 \text{ m}$  apart, what is the new force?
3. Two identical small charges are  $0.30 \text{ m}$  apart and repel with a force of  $0.040 \text{ N}$ . What is the charge on each object?
4. A  $+5.0 \mu\text{C}$  charge and a  $-2.0 \mu\text{C}$  charge attract with a  $1.0 \text{ N}$  force. How far apart are their centers?
5. In a hydrogen atom, the electron and proton are about  $5.3 \times 10^{-11} \text{ m}$  apart. What is the electric force between them? What acceleration does this give the electron? (electron mass =  $9.11 \times 10^{-31} \text{ kg}$ )

6. What is the electric field 0.50 m east of a +8.0 nC charge?

7. How far from a +2.0 nC charge is the electric field strength 450 N/C?

8. A point charge makes a  $3.6 \times 10^4$  N/C electric field at a location 5.0 cm away. What is the magnitude of the charge?

9. A -3.0  $\mu\text{C}$  charge is placed in a uniform 250 N/C electric field pointing right. What force acts on the charge?

10. A +3.0  $\mu\text{C}$  bead is 0.20 m from another charged bead. What is the electric force between them?

11. A 0.20 g particle with charge +50 nC is in a uniform 4000 N/C electric field pointing right. What is the particle's acceleration if no other horizontal forces act on it?

12. A 2.0 g bead with charge  $+0.50 \mu\text{C}$  starts from rest in a uniform 300 N/C field pointing right. If no other horizontal force acts on it, how fast is it moving after 0.40 s?

13. A  $+2.0 \text{ nC}$  charge is 10 cm to the left of a  $-2.0 \text{ nC}$  charge. What is the electric field at the midpoint between them?

14. A  $+3.0 \text{ nC}$  charge is at  $x = 0 \text{ cm}$ , and a  $-1.0 \text{ nC}$  charge is at  $x = 20 \text{ cm}$ . What is the electric field at  $x = 5 \text{ cm}$ ?

15. In fair weather, Earth's atmosphere has an electric field of about 100 N/C pointing down. What force does this field put on an electron? Compare it to the electron's weight. (electron mass =  $9.11 \times 10^{-31} \text{ kg}$ )

16. A  $+6.0 \text{ nC}$  source charge is 15 cm west of a  $-2.0 \text{ nC}$  test charge. What force acts on the test charge?

17. Three charges are on a line. A  $+4.0\ \mu\text{C}$  charge is at  $x = 0\ \text{cm}$ , a  $+2.0\ \mu\text{C}$  charge is at  $x = 30\ \text{cm}$ , and a  $-3.0\ \mu\text{C}$  charge is at  $x = 50\ \text{cm}$ . What is the net force on the  $+2.0\ \mu\text{C}$  charge?

18. Three charges are on a line. A  $-2.0\ \mu\text{C}$  charge is at  $x = 0\ \text{cm}$ , a  $+1.0\ \mu\text{C}$  charge is at  $x = 20\ \text{cm}$ , and a  $+3.0\ \mu\text{C}$  charge is at  $x = 50\ \text{cm}$ . What is the net force on the  $+1.0\ \mu\text{C}$  charge?

19. A  $+10\ \text{nC}$  charge is  $20\ \text{cm}$  west of a point, and a  $-5.0\ \text{nC}$  charge is  $10\ \text{cm}$  east of the same point. A  $-2.0\ \mu\text{C}$  bead with mass  $3.0\ \text{g}$  is placed at the point. What is the bead's acceleration?

20. A  $+4.0\ \text{nC}$  charge is at  $x = 0\ \text{cm}$  and a  $+1.0\ \text{nC}$  charge is at  $x = 30\ \text{cm}$ . Where between the charges is the electric field zero?

21. A  $3.0\ \text{g}$  pith ball with charge  $-20\ \text{nC}$  hangs near a charged board. Model the nearby part of the board as having  $+35\ \text{nC}$  of charge at a point  $2.0\ \text{cm}$  away. What is the electric force, and is it larger than the ball's weight?